

|          | Problems                                                                                                                                                                                                                         | Solutions                                                                                                                                                                                                                                                                            |
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| Sprint 1 | Initially, we were not planning to use the motor controller, or the battery, therefore the design changed several times.                                                                                                         | The design was changed, and we came up with integrating them on lid, so design of it was changed, and printed accordingly                                                                                                                                                            |
|          | The robot's height wasn't planned, and there was less space for the wires inside.                                                                                                                                                | The robot height was adjusted and the body was reprinted with the correct height.                                                                                                                                                                                                    |
|          | The wing had less width, and therefore was fragile.                                                                                                                                                                              | The wing width was fixed, and made sure it was strong.                                                                                                                                                                                                                               |
|          | Legs were initially bigger, causing friction.                                                                                                                                                                                    | The size of the legs was decreased and printed                                                                                                                                                                                                                                       |
| Sprint 2 | The locations of the ultrasonic sensors, wings, light sensor were changed due to space management.                                                                                                                               | The locations were planned accordingly, space was managed, and the required CAD parts were printed on time.                                                                                                                                                                          |
|          | The power for the servos together wasn't enough                                                                                                                                                                                  | A separate power supply was used for servos, which was power supply from batteries.                                                                                                                                                                                                  |
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| Sprint 3 | The connection of the wing and the body was challenging since there was slight place in the wing connection.                                                                                                                     | We reduced the complexity by using equipments to handle the bolts.                                                                                                                                                                                                                   |
|          | The servo bolts were too small compared to the holes printed.                                                                                                                                                                    | The materials were reprinted with smaller holes that fit into the bolt.                                                                                                                                                                                                              |
| Sprint 4 | The robot was tested several times, as the wing motor started glitching, since the angles for it were wrong. Next, I tested the code several times, for changing the threshold values and finding the best values for the robot. | The wing motor was replaced with the extra motor we bought, as the motor was burnt and not functioning well, and the angles were premeasured and adsuted before it was integrated with the wing. To decide the threshold values, the test was carried out several times and decided. |
|          | The robot initially had an issue with moving slow, since there was larger friction. Once it was solved, there was a problem with the turnings, where the robot didn't make clear turns.                                          | As a solution for the drag issue, the wave gate movement was shifted to crawl gate movement. The clear turns were made by finetuning the threshold values.                                                                                                                           |
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| Sprint 5 | The robot is confusing the detections of obstacle and predator, since predator has the higher priority, obstacle avoidance doesn't override even when there is an obstacle. | A subsumption is made in a way that both works.                                                                                      |
|          | Integration wasn't successful, as the priorities didn't work as expected                                                                                                    | Subsumption architecture was changed as the previous code had some issues. The new code works perfectly according to the priorities. |